Mock AIO 2026

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Practice for AIO, but no test data provided, so just think, no implementation. The difficulty is probably around the absolute worse you'll get in AIO.

For some reason so many problems involve an array and an $O(N)$ solution.

Measures of Center

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 32 MB **Output File:** `Standard Output`

You are given an array/list/vector of integers. Find the subarray with the maximum sum of it's median + mean.

A subarray of an array is formed by deleting 0 or more elements from the start of the array, and deleting 0 or more elements from the end of the array.
The mean of an array is the sum of the array divided by the number of numbers/elements in the array.
The median of an array is the middle element of the array once sorted. If there are an even number of elements, the median is the mean of the 2 middle elements of the array.

Input Format

The first line of input contains the integer n, the length of the array of integers. The next line of input contains n space-separated integers, the elements of the array.

Output Format

You should output 2 space-separated integers a and b, the starting and ending index of the desired subarray (0-indexed).

Subtasks & Constraints

For all subtasks, $n \geq 0$, and all elements of the array are between -10^9 and 10^9.

Subtask	Points	Additional Constraints	Description
1	10	$N \leq 10^2$	
2	30	$N \leq 10^3$	
3	60	$N \leq 10^6$	

Enclosed Mode

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are given an array of n integers. Your task is to find the length of the smallest subarray that includes all numbers of a value with the highest frequency in the array.

Input Format

The first line of input is n, and the next line contains n space-separated integers, the elements of the array.

Output Format

Output the length of the smallest subarray.

Subtasks & Constraints

For all test cases, $n \geq 0$, and all elements of the array are between -10^9 and 10^9.

Subtask	Points	Additional Constraints	Description
1	30	$n \leq 10^3$	
2	70	$n \leq 10^6$	

Alternating Sum

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 64 MB **Output File:** `Standard Output`

You are given an array of length n of integers. Find the/a subarray $(b_1, b_2, \ldots, b_k)$ which has the maximum sum of $b_1 - b_2 + b_3 \ldots (-1)^{i+1} b_i$ (alternating positive and negative).

Input Format

The first line of input contains the integer n. The next line of input contains n space-separated integers, the integers in the array.

Output Format

Output the maximum alternating sum of any subarray.

Subtasks & Constraints

For all subtasks, $n \geq 0$ and the elements of the array are between -10^3 and 10^3.

Subtask	Points	Additional Constraints	Description
1	15	$n \leq 10^2$	
2	35	$n \leq 10^3$	
3	50	$n \leq 10^6$	

Interception

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 256 MB **Output File:** `Standard Output`

TheGOAT has run away with a USB with leaked camp selection exams, and the AIOC desparately needs to catch him before they have to remake 50 problems for camp. The AIOC knows where TheGOAT will be at every point of time from 1 to t.

Given the location of camp (where everyone from AIOC will be), and a graph of all places people from AIOC can travel (1 edge per time unit), what is the earliest time when AIOC can capture TheGOAT to prevent catastrophe (AIOC is at the same place as TheGOAT at that specific point of time)?

Input Format

The first line of input contains 4 space-separated integers: t, s (the position of camp), v (the number of places there are), and e (the number of connections between places). The next e lines of input contain 2 space-separated integers a_i and b_i, meaning there is a bidirectional path between place a_i and b_i.

Output Format

Output a single integer, the earliest time unit when the AIOC can capture/be on the same place as TheGOAT. If AIOC can never catch TheGOAT before time t, print "Disaster".

Subtasks & Constraints

For all test cases, $0 \leq t, s, v, e, a_i, b_i$, no path will be mentioned twice. Sorry for this problem I don't have good subtasks.

Subtask	Points	Additional Constraints	Description
1	20	$t, v, e \leq 10^2$	
2	30	$t, v, e \leq 10^3$	
3	50	$t, v, e \leq 10^6$	

Skipping the queue

Time Limit: 0.5 second	**Input File:** `Standard Input`
Memory Limit: 128 MB	**Output File:** `Standard Output`

You are in a long queue to get the newly released GTA 7. However, you are planning to spend a lot of money on *buying it*. Therefore, at any position in the line, you can coerce a person less that or equal to k places in front of you to give up their spot for a given cost. Of course, the person in front of you won't let you off, and you'll need to bribe them as well to be able to move.

Input Format

The first line of input contains 2 space-separated integers n, the number of people in the queue, and k. The second line of input contains n space separated-integers $a_1, a_2, \ldots, a_n$, the cost of coercing a person to swap places. The third line of input contains n space separated-inters $b_1, b_2, \ldots, b_n$, the cost of bribing a person.

Output Format

You should output a single integer, the minimum amount of money you have to spend to go from the back of the queue (before position 1) to the end of the queue (after position n).

Subtasks & Constraints

For all test cases, $n, k \geq 0$, and $0 \leq a_i, b_i \leq 10^6$

Subtask	Points	Additional Constraints	Description
1	30	$N \leq 10^5$	Supposed to be a $O(N \log N)$
2	70	$N \leq 10^7$	Supposed to be $O(N)$

Note

If you are not using python, you WILL have to use 64 bit-integers, unless your code will fail on larger test cases.

Also, c++/c users can probably pass an $O(N \log N)$ solution even with 10^7 by vectorising or just constant optimisations

Atlantis IV: The lost city

Time Limit: 1 second **Input File:** `Standard Input`
Memory Limit: 256 MB **Output File:** `Standard Output`

After years of searching, an ancient scroll has been recovered from the city of Atlantis! It is well known that Atlantis has 2 strips of land separated by a river in the middle of it, with points 0, 1, 2, $\ldots, k$ on each strip. It has also been discovered that there were 2 bridges, with length 1, connecting the strips of land in atlantis, namely from point 0 from the first strip to second strip, and a similar bridge at point k.

Your goal is to find the most common trips from a point on the first strip of land to a point on the second strip of land, using the information given on the scroll. The scroll contains k integers $a_1, a_2, \ldots, a_k$. For each i, a_i is the sum of the fastest route to complete a trip from a point on the first strip to the second strip. *The trips outlined in the scroll are the commont trips you want to find.* Any valid list of trips will be accepted.

Input Format

The first line of input contains the integer k. The next line of input contains k space-separated integers $a_1, a_2, \ldots, a_k$.

Output Format

If you have found n trips, you should output n lines with 2 space-separated integers s_i and e_i, the start and end point of each trip.

Subtasks & Constraints

For all subtasks, $0 \le k \le 10^6$, and for all i, $0 \le a_i \le 10^9$. It is also guaranteed that none of the trips start or end at the same point.

Subtask	Points	Additional Constraints	Description
1	30		There are either 0 or 1 trip(s)
2	30	$k \le 10^3$	
3	40		Full task